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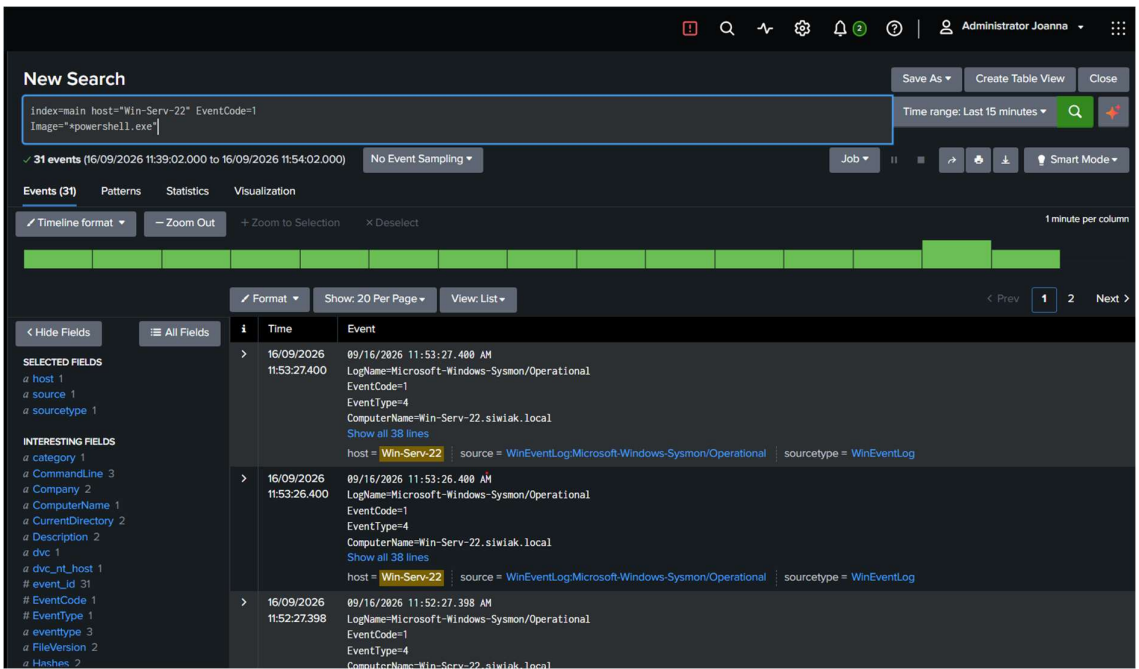
16/09/2026

This phase focuses on developing and validating a Splunk detection for suspicious PowerShell activity, mapped to MITRE ATT&CK technique T1059.001 – PowerShell. Sysmon Process Creation telemetry (Event ID 1) is used to establish a baseline of legitimate PowerShell activity before generating controlled suspicious activity and developing the detection logic.

Searches

SPL 1- Identify PowerShell process activity

The initial search identified Sysmon Event ID 1 events associated with PowerShell processes on Win-Serv-22. The search returned both normal user PowerShell activity and processes generated by the Splunk Universal Forwarder.



SPL 2- Review PowerShell process details

The results were formatted to examine the user, executable path, command line and parent process. This revealed frequent splunk-powershell.exe activity generated legitimately by the Splunk Universal Forwarder, demonstrating a potential source of false positives.

New Search Save As Create Table View Close

index=main host="Win-Serv-22" EventCode=1 Image="powershell.exe"
 | table _time User Image CommandLine ParentImage
 | sort - _time

Time range: Last 15 minutes Q +

✓ 31 events (16/09/2026 11:40:38.000 to 16/09/2026 11:55:38.000) No Event Sampling Job || → ↺ ↻ ⬇ Smart Mode

Events Patterns **Statistics (31)** Visualization

Show: 20 Per Page Format Preview: On < Prev 1 2 Next >

_time	User	Image	CommandLine	ParentImage
2026-09-16 11:55:27.401	NOT_TRANSLATED NT AUTHORITY\SYSTEM	C:\Program Files\SplunkUniversalForwarder\bin\splunk-powershell.exe	"C:\Program Files\SplunkUniversalForwarder\bin\splunk-powershell.exe" --ps2	C:\Program Files\SplunkUniversalForwarder\bin\...
2026-09-16 11:55:26.401	NOT_TRANSLATED NT AUTHORITY\SYSTEM	C:\Program Files\SplunkUniversalForwarder\bin\splunk-powershell.exe	"C:\Program Files\SplunkUniversalForwarder\bin\splunk-powershell.exe"	C:\Program Files\SplunkUniversalForwarder\bin\...
2026-09-16 11:54:27.401	NOT_TRANSLATED NT AUTHORITY\SYSTEM	C:\Program Files\SplunkUniversalForwarder\bin\splunk-powershell.exe	"C:\Program Files\SplunkUniversalForwarder\bin\splunk-powershell.exe" --ps2	C:\Program Files\SplunkUniversalForwarder\bin\...
2026-09-16 11:54:26.401	NOT_TRANSLATED NT AUTHORITY\SYSTEM	C:\Program Files\SplunkUniversalForwarder\bin\splunk-powershell.exe	"C:\Program Files\SplunkUniversalForwarder\bin\splunk-powershell.exe"	C:\Program Files\SplunkUniversalForwarder\bin\...
2026-09-16 11:53:27.400	NOT_TRANSLATED NT AUTHORITY\SYSTEM	C:\Program Files\SplunkUniversalForwarder\bin\splunk-powershell.exe	"C:\Program Files\SplunkUniversalForwarder\bin\splunk-powershell.exe" --ps2	C:\Program Files\SplunkUniversalForwarder\bin\...
2026-09-16 11:53:26.400	NOT_TRANSLATED NT AUTHORITY\SYSTEM	C:\Program Files\SplunkUniversalForwarder\bin\splunk-powershell.exe	"C:\Program Files\SplunkUniversalForwarder\bin\splunk-powershell.exe"	C:\Program Files\SplunkUniversalForwarder\bin\...
2026-09-16 11:52:27.398	NOT_TRANSLATED NT AUTHORITY\SYSTEM	C:\Program Files\SplunkUniversalForwarder\bin\splunk-powershell.exe	"C:\Program Files\SplunkUniversalForwarder\bin\splunk-powershell.exe" --ps2	C:\Program Files\SplunkUniversalForwarder\bin\...
2026-09-16 11:52:26.398	NOT_TRANSLATED NT AUTHORITY\SYSTEM	C:\Program Files\SplunkUniversalForwarder\bin\splunk-powershell.exe	"C:\Program Files\SplunkUniversalForwarder\bin\splunk-powershell.exe"	C:\Program Files\SplunkUniversalForwarder\bin\...

SPL 3- Establish legitimate PowerShell baseline

The Splunk Universal Forwarder activity was excluded to isolate interactive PowerShell execution. The remaining event showed PowerShell launched by the user from explorer.exe without suspicious command-line arguments, establishing a baseline for normal PowerShell activity.

New Search Save As Create Table View Close

index=main host="Win-Serv-22" EventCode=1
 Image="powershell.exe"
 NOT Image="*SplunkUniversalForwarder*"
 | table _time User Image CommandLine ParentImage
 | sort - _time index=main host="Win-Serv-22" EventCode=1

Time range: Last 15 minutes Q +

✓ 1 event (16/09/2026 11:42:01.000 to 16/09/2026 11:57:01.000) No Event Sampling Job || → ↺ ↻ ⬇ Smart Mode

Events Patterns **Statistics (1)** Visualization

Show: 20 Per Page Format Preview: On < Prev 1 2 Next >

_time	User	Image	CommandLine	ParentImage
2026-09-16 11:52:01.453	NOT_TRANSLATED SIMIAK\jakub	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe	"C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe"	C:\Windows\explorer.exe

Controlled suspicious PowerShell activity

A controlled PowerShell test was performed using the -EncodedCommand parameter. The Base64-encoded payload contained only a benign Write-Output "DetectionLab-Test" command, allowing suspicious command-line behaviour to be simulated safely.

```
PS C:\Users\jakub> $cmd = 'Write-Output "DetectionLab-Test" '
PS C:\Users\jakub> $encoded = [Convert]::ToBase64String([Text.Encoding]::Unicode.GetBytes($cmd))
PS C:\Users\jakub> $encoded
VwByAGkAdAB1AC0ATwB1AHQAcAB1AHQAIAAIAEQAZQB0AGUAYwB0AGkAbwBuAEwAYQB1AC0AVAB1AHMAdAAIACAA
PS C:\Users\jakub> powershell.exe -EncodedCommand $encoded
DetectionLab-Test
PS C:\Users\jakub>
```

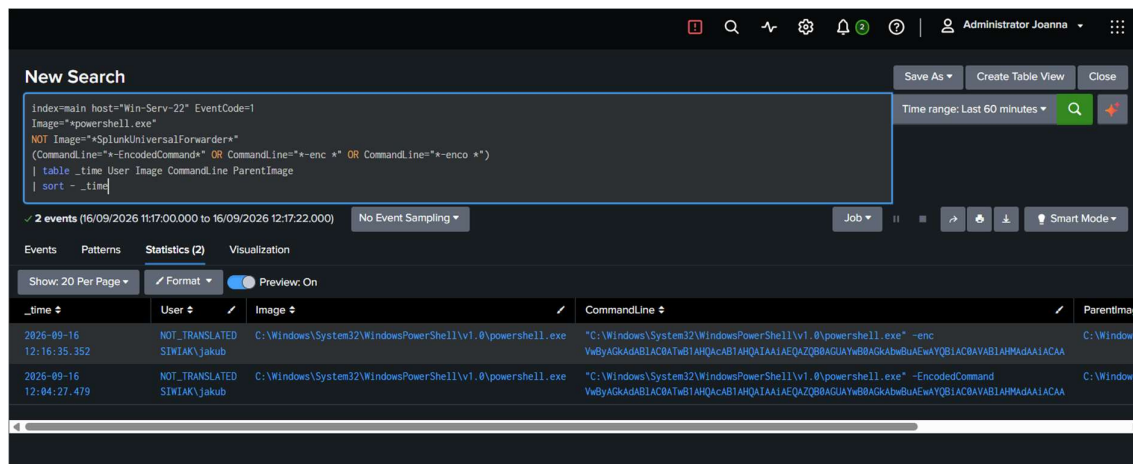
The controlled test was successfully identified in Splunk using Sysmon Event ID 1 telemetry. The event captured the PowerShell executable, executing user and the -EncodedCommand parameter with the Base64-encoded payload. This confirmed that the required telemetry was available for detecting encoded PowerShell execution.

The screenshot shows the Splunk search interface with a search query: `index=main host="Win-Serv-22" EventCode=1 Image="powershell.exe" CommandLine="*-EncodedCommand*" NOT Image="*SplunkUniversalForwarder*"`. The results table shows a single event from 2026-09-16 at 12:04:27.479, executed by user SIWIAK\jakub. The command line is `"C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe" -EncodedCommand VwByAGkAdAB1AC0ATwB1AHQAcAB1AHQAIAAIAEQAZQB0AGUAYwB0AGkAbwBuAEwAYQB1AC0AVAB1AHMAdAAIACAA`, and the parent image is `C:\Windows`.

_time	User	Image	CommandLine	ParentImage
2026-09-16 12:04:27.479	SIWIAK\jakub	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe	"C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe" -EncodedCommand VwByAGkAdAB1AC0ATwB1AHQAcAB1AHQAIAAIAEQAZQB0AGUAYwB0AGkAbwBuAEwAYQB1AC0AVAB1AHMAdAAIACAA	C:\Windows

The detection was validated against two PowerShell encoded-command variants: -EncodedCommand and the abbreviated -enc parameter. Both controlled executions were successfully identified using Sysmon Event ID 1 telemetry, confirming that the detection is not limited to a single command-line representation.

```
DetectionLab-Test
PS C:\Users\jakub> powershell.exe -enc $encoded
DetectionLab-Test
PS C:\Users\jakub>
```



Detection

A scheduled Splunk alert named Suspicious PowerShell Encoded Command - T1059.001 was created to continuously monitor the Windows Server endpoint for encoded PowerShell execution. The detection uses Sysmon Event ID 1 (Process Creation) telemetry and examines the PowerShell command line for encoded-command parameters such as -EncodedCommand, -enc, and -enco.

The alert runs every five minutes and analyses events from the previous five-minute period. It triggers when at least one matching event is identified and records the detection in Splunk Triggered Alerts with Medium severity. Legitimate splunk-powershell.exe activity generated by the Splunk Universal Forwarder is excluded to reduce false positives.

The detection is mapped to MITRE ATT&CK T1059.001 – PowerShell. Encoded PowerShell is not inherently malicious, therefore a triggered alert should be treated as an indicator requiring further investigation rather than confirmation of compromise.

Suspicious PowerShell Encoded Command - T1059.001

Detects encoded PowerShell execution using Sysmon Event ID 1, mapped to MITRE ATT&CK T1059.001.

Enabled: Yes. [Disable](#)

App: search

Permissions: Shared in App. Owned by admin. [Edit](#)

Modified: 16 Sept 2026 12:25:01

Alert Type: Scheduled. Cron Schedule. [Edit](#)

Trigger Condition: .. Number of Results is > 0. [Edit](#)

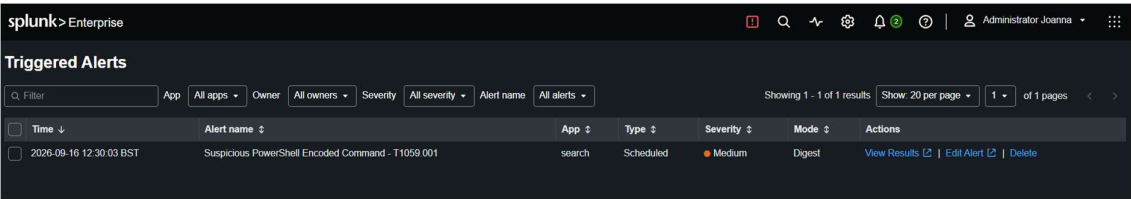
Actions: [▼](#) 1 Action [Edit](#)

[🔍](#) Add to Triggered Alerts

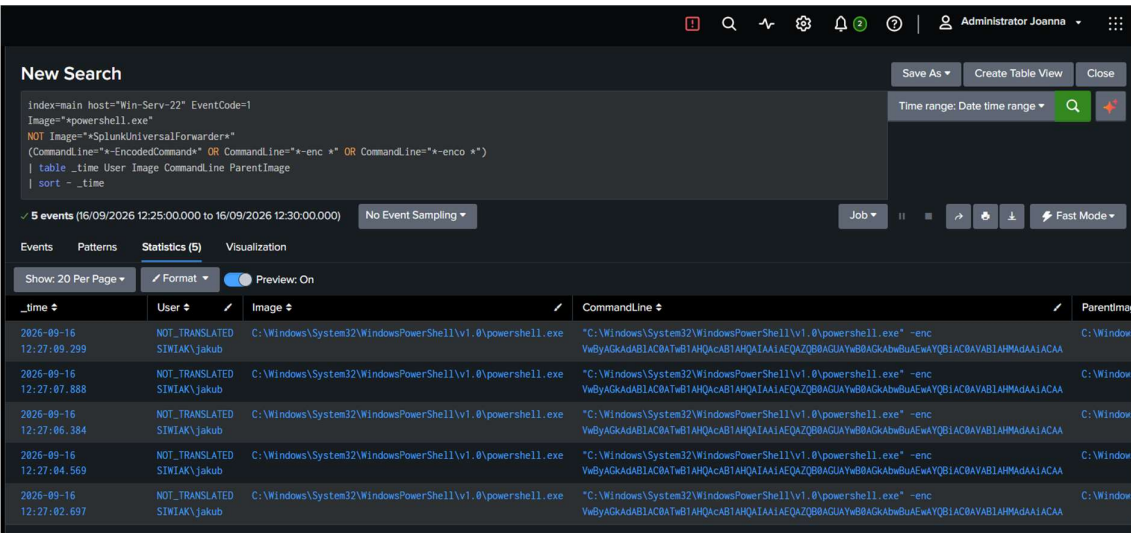
Final Validation

The deployed detection was successfully validated end-to-end. Multiple controlled PowerShell executions using the abbreviated -enc parameter were generated on Win-Serv-22. Sysmon captured the process creation events and the telemetry was forwarded to Splunk.

During its scheduled five-minute search window, the detection identified five matching events and triggered the Suspicious PowerShell Encoded Command - T1059.001 alert with Medium severity. The alert was successfully recorded in Splunk Triggered Alerts, confirming that the complete detection pipeline was operational.



Triggered Alert



Detection Results

False Positives and Limitations

Encoded PowerShell is not inherently malicious. Legitimate administrative scripts, automation tools and management software may use encoded commands, therefore a detection should be investigated before being treated as a confirmed security incident.

The current detection focuses specifically on encoded PowerShell execution and does not detect every form of malicious PowerShell activity. Additional detections would be required for other suspicious behaviours such as download commands, execution-policy bypasses or unusual PowerShell parent-child process relationships.

During baseline analysis, legitimate splunk-powershell.exe activity generated by the Splunk Universal Forwarder was identified and excluded from the detection to reduce false positives.

MITRE ATT&CK Mapping

Technique: T1059.001 – PowerShell

Tactic: Execution

Data Source: Sysmon Event ID 1 – Process Creation

Detection focus: PowerShell processes containing encoded-command parameters such as -EncodedCommand, -enc or -enco.

Conclusion

The detection was successfully developed and validated using controlled PowerShell activity. Sysmon provided the required process creation telemetry, Splunk identified multiple encoded-command variants, and the scheduled detection successfully generated a Medium-severity triggered alert.

This phase demonstrates the complete detection engineering workflow from baseline analysis and false-positive identification through detection development, controlled testing and final alert validation.